北京郵電大學

Queen Mary University of London

EBU6408
Mock Exam

Joint Programme Examinations 2023/24

EBU6408 Sensors and Radio Frequency Identification

Mock Exam

Time allowed 2 hours

Answer ALL questions

For examiners' use only

1	
2	
3	
4	
5	
6	
7	
8	
Total	

Complete the information below about yourself very carefully.

QM student number	
BUPT student number	
Class number	

NOT allowed: Any Electronic devices including electronic calculators and electronic dictionaries.

INSTRUCTIONS

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Fatma Benkhelifa, Dr Jin Zhang

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Filename: 2324_EBU6408_Mock_Exam No answer book required

Instructions

Before the start of the examination

1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.

2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance**. **Possession of mobile phones, electronic devices and unauthorised materials is an offence.**

3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**

4) Do not turn over your question paper or begin writing until told to do.

During the examination

1) You must not communicate with or copy from another student.

2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.

3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.

4) For 2 hour examinations you may **not** leave temporarily.

5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**

2) Remain in your seat until you are told you may leave.

Question 1

a) Automatic identification systems:

[10 marks]

i) What are the different types of automatic identification systems?

(3 marks)

ii) Describe three of them.

(3 marks)

iii) Describe the different steps of face recognition:

(2.5 marks)

iv) What are the technical features of speech recognition?

(1.5 marks)

b) Barcode and RFID are among the identification technologies that are being used in the enterprise environment.

[15 marks]

i) Define the radio frequency identification. To which era dates back RFID?

(4 marks)

ii) State four main features of RFID system? Describe them.

(4 marks)

iii) What are the main components of an RFID system?

(3 marks)

i) What are the main differences between 1D and 2D barcode recognitions?

(4 marks)

Question 2

a) RFID middleware:

[4 marks]

i) Define the RFID middleware.

(1 mark)

ii) What is its main function?

(3 marks)

b) RFID Tags

[7 marks]

i) Define the tag antenna.

(3 marks)

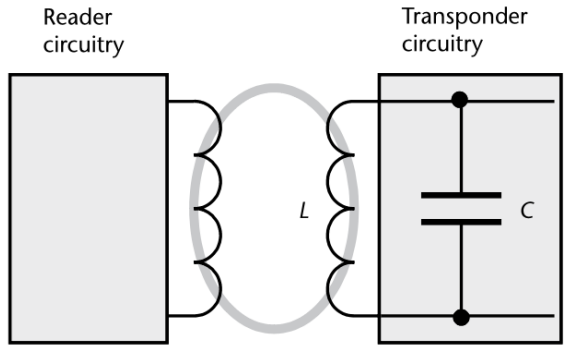
ii) Explain the different tag antenna types.

(2 marks)

iii) How the coil antenna works?

(2 marks)

c) The reader produces a magnetic field that triggers the tag as shown in Figure. When the reader receives the transmitted data, it interprets the data and takes appropriate action. When the transponder enters the field produced by the reader, the coil produces a voltage inside the tag. In a passive transponder, this voltage can be used to power the tag. In an active transponder, the voltage is used to wake the tag and use its internal battery.



RFID system and a resonant frequency calculation.

- Active transponders generally have longer read distances and shorter operational life and are larger and more costly to manufacture.
- Passive transponders are generally smaller, have a longer life, and are less expensive to manufacture. For optimum performance, the transponder coil is used in a parallel LC circuit designed to resonate at the operating frequency of the reader.

[8 marks]

i) Calculate the capacitor value for a 4.9-mH transponder coil operating at 125 kHz.

(2 marks)

ii) What would be the resonant frequency f_1 if the overlapped tags have a new total inductance of 5.5 mH?

(2 marks)

iii) Formula for EIRP in Watts is as follows:

$$EIRP = \frac{E^2 r^2}{30}$$

where *E* is in V/m, and *r* is distance in meters. Derive the EIRP in dBm, using *E* in dBμV/m and *r* in meters.

(2 marks)

iv) In standard test setups, the electrical field strength is often measured at a distance of 3m. Show that in this case we can use the simplified formula:

(2 marks)

d) You want a RFID tag that supports longer distance communications, rely on the reader to provide power to the tag, and cannot be written.

[6 marks]

i) What kind of tag(s) do you need?

(3 marks)

ii) Discuss advantages and disadvantages of these tags.

(3 marks)

Question 3

In modelling a real situation with a reader, many tags and other objects in between is an important part of the RFID system design, and the use of RF simulation tools in all parts of the system helps to predict the range of reliable operation.

Typical UHF operating parameters are as follows:

- Reader transmit power: $P_R = 33$ dBm (2W).
- System operating wavelength: $\lambda = 0.33$m (915 MHz).
- Reader receiver sensitivity: $S_R = -80$ dBm.
- Reader antenna gain: $G_R = 4$ (6 dBi).
- Tag power (sensitivity) requirement: $P_T = -10$ dBm.
- Tag antenna gain: $G_T = 1.26$ (1 dBi);
- Tag backscatter efficiency: $E_T = 0.01$ or 1% (–20 dB) calculated as $(\Delta\rho)^2$, where $\Delta\rho$ is the differential reflection coefficient.

Take into consideration the following sources of losses: the cables and connectors losses at the reader (-3dB), polarization mismatches at the reader antenna (-3dB), and tag antenna matching loss (-1dBi)

a) Forward link budget: The signal received at the tag has to be bigger or, in the worst case, equal to the tag sensitivity threshold. In case the tag is at the far edge of the interrogation zone, we can say that:

[12 marks]

i) Explain the sources of energy attenuation in the forward link budget.

(3 marks)

ii) Calculate the maximum free-space loss (FSL) allowed to meet the tag sensitivity threshold.

(3 marks)

iii) Calculate the maximum distance between the tag and reader at 915MHz.

(3 marks)

b) Reverse link budget: The signal received at the tag has to be bigger or, in the worst case, equal to the tag sensitivity threshold. In case the tag is at the far edge of the interrogation zone, we can say that:

[8 marks]

i) Explain the sources of energy attenuation in the reverse link budget.

(3 marks)

ii) Calculate the reader received power.

(3 marks)

iii) Has the reader met the sensitivity threshold? What is the signal margin at the reader's receiver?

(2 marks)

c) Differential Bi-Phase (DBS) RZ coding:

[5 marks]

i) Explain the principle of unipolar RZ coding.

(2 marks)

ii) What is the unipolar RZ coding of the following binary sequence:

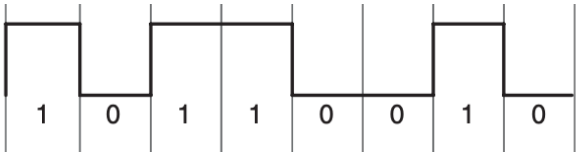


Figure 2

(3 marks)

Question 4

a) We have four transponders in the interrogation zone of the reader. The transponders are uniquely identified using 8-bit serial numbers in the range 00-FFh (=0-255 dec or 00000000 – 11111111 bin).

Tag 1	10110010
Tag 2	10100011
Tag 3	10110011
Tag 4	11100011

The reader uses the **binary search based anti-collision algorithm**. In one of the iterations, following the transmission of the command Request (<11111111) by the reader. As a result, a collision occurs.

[10 marks]

i) Explain the reason of the collision. How many possibilities for the serial numbers?

(3 mark)

ii) To deal with this collision, what is the next request command that the reader can perform?

(2 mark)

iii) Assume the reader sends a REQUEST command (<10111111), what will happen? Does a collision will occur? At which bits a collision might occur?

(3 mark)

iv) If a collision occurs, how many possibilities we have? List them down.

(1 mark)

v) Assume eight transponders are in the interrogation area of the reader. What is the average number of iterations to detect a single tag using the binary search based anti-collision algorithm?

(1 mark)

b) EPC Class 0 standard:

[7 marks]

vi) When did EPC started? What was its goal?

(1 mark)

vii) State at least two universities that participate in the EPCglobal partnership.

(1 mark)

viii) When and who founded EPCglobal Network? What was its mission?

(1 mark)

ix) Draw a diagram of a typical EPCglobal network and explain two of its five basic services.

(2 marks)

x) For which types of tags stands EPCglobal class 0 protocol? How many bits are included in EPC code? Which anti-collision protocol is used in this standard?

(2 marks)

d) RFID applications and practice:

[8 marks]

i) State three patterns of RFID applications and explain them.

(4 marks)

iii) State at least three types of attacks to transponders.

(3 marks)

iv) State two applications for RFID information retrieval and integration.

(2 marks)